Akoizo Original Practice - Form 4 - Answer key

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) What gas inflates the balloon in the vinegar and baking soda action?

A. Hydrogen

B. Nitrogen

C. Carbon dioxide

D. Oxygen

Correct: C

2. (2 points) Which property most cleanly distinguishes a quarter from a nickel?

A. Color

B. Magnetism

C. Diameter

D. Mass

Correct: C

3. (2 points) Action 6 starts before action 5 on the ASL. What happens to action 5?

A. It earns zero points, even if it completes later

B. It still earns 50

C. It earns 25

D. The run restarts

Correct: A

4. (2 points) Which component is NOT allowed?

Unit: Electrical, infrared, and magnetic actions.
Worked-case data: A 9 V battery drives a nichrome segment of 6 Ω. What current and power result, and is the plan legal?
Separate investigation observations:
Timeline: Coin start at 0 s; IR action at 8 s; nichrome cut at 27 s.
Slow test day: Device ran 15 percent slower; cut happened at 31 s.
Supervisor note: Electricity used after 30 s.

A. A microcontroller

B. A relay

C. An IR LED

D. A 9 V battery

Correct: A

5. (4 points) With a pulley IMA of 3, how far must the effort rope move to lift a load 12 cm?

answer: 36 cm, three times the load’s rise. (4 points).

6. (4 points) Calculate the size points for a device measuring 70.0 x 45.0 x 79.5 cm.

answer: 10 cm under -> 10 points; 35 cm under -> capped at 30; 0.5 cm under -> 0.5. Total 40.5. (4 points).

7. (4 points) Why aim slightly under the Target Time?

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

answer: Normal variation may push the run longer; going over costs 2 points per second, while finishing slightly early loses only a few points. (4 points).

8. (4 points) Why should reactants be pre-measured and labeled?

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

answer: It makes the reaction consistent, speeds setup, and ensures safe, known contents for inspection. (4 points).

9. (4 points) Why should the first action be clearly visible?

Unit: The Start Action: coins, slots, and selective triggers.
Worked-case data: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.
Separate investigation observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

answer: Timing starts with the first action, so the supervisor must see it to start the clock accurately. (4 points).

10. (4 points) How can a team earn the 75-point setup bonus at State?

answer: By completing setup within 15 minutes, which requires practiced routines and pre-measured, labeled supplies. (4 points).

11. (6 points) Solve this case and explain your reasoning: A 9 V battery drives a nichrome segment of 6 Ω. What current and power result, and is the plan legal?

reasoning: Current I = V ÷ R = 9 ÷ 6 = 1.5 A.
Power P = V x I = 9 x 1.5 = 13.5 W, enough to melt nylon line quickly.
Legal if it uses one 9 V labeled battery and finishes within the first 30 s. (3 points).

conclusion: The hot wire works fast; schedule it early and disconnect power after the line breaks. (3 points).

12. (2 points) Does the pulley action qualify?

Supplied synthetic observations:
Pulley build: Load 420 g lifts 11 cm; effort rope moves 22 cm.
Supervisor check: Counts two supporting segments.
Team claim: Pulley action meets IMA 3.

A. Yes; it lifts more than 10 cm

B. No; IMA is only 2

C. Yes; the load exceeds 400 g

Correct: B

13. (2 points) Define “Start Action” in the context of Mission Possible.

answer: Coin-triggered first action, worth 25, 50, or 200 points. (2 points).

14. (2 points) Define “Target Operation Time” in the context of Mission Possible.

answer: The run length teams aim for, announced at setup. (2 points).

15. (2 points) Define “Acetic acid” in the context of Mission Possible.

answer: Acid in vinegar, about 5 percent in household vinegar. (2 points).

16. (2 points) Define “Effort arm” in the context of Mission Possible.

answer: Distance from fulcrum to where the effort is applied. (2 points).

17. (2 points) Define “First action” in the context of Mission Possible.

answer: Visible motion that starts timing. (2 points).

18. (2 points) Define “Dead-end action” in the context of Mission Possible.

answer: An action that does not lead to the Final Action; scores nothing. (2 points).